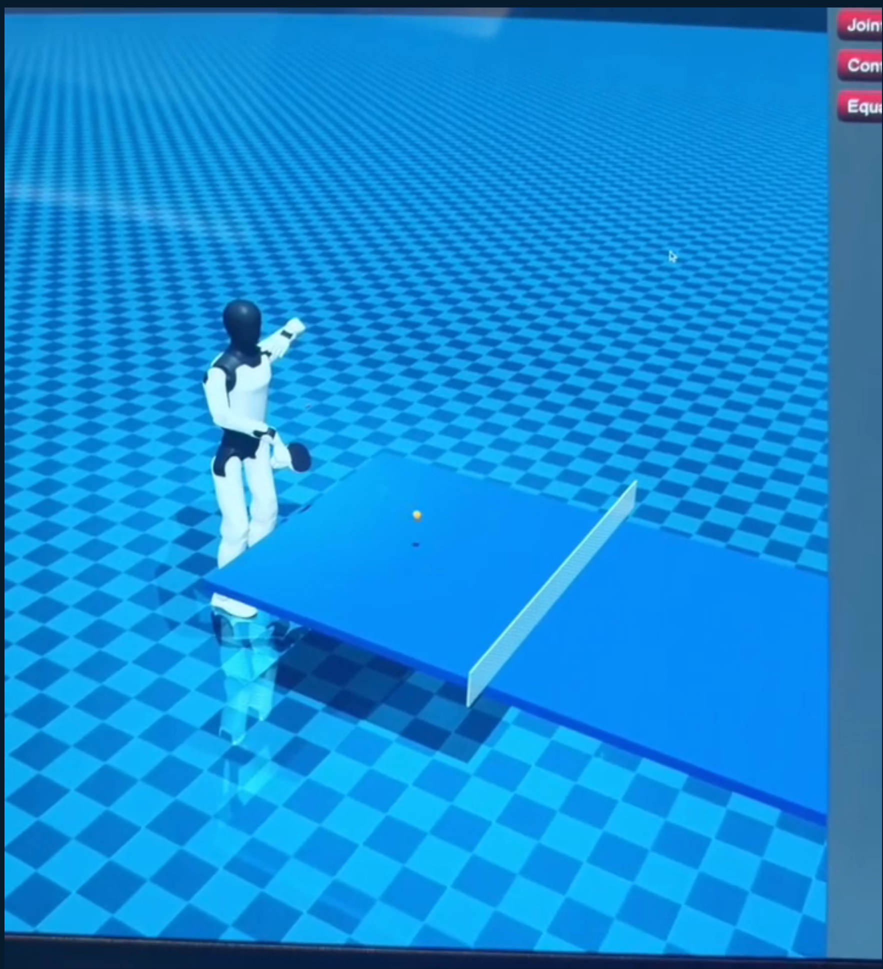
HOPE / OPEN-SOURCE HUMANOID PING-PONG

A3 MuJoCo Serve Package

High-level build, qualification, and on-robot execution guide

PR #18 reference: hardware-tested / fixed-profile approved / deployable



Visual reference from the supplied 6.525-second PR #18 demo. This guide uses only the exact machine-validated PR #18 motion.

HOPE nightly_built / 31 July 2026

01 / PACKAGE PURPOSE

One approved motion, from package to A3 execution

This operator walkthrough treats the original PR #18 CSV as immutable and follows it through qualification, native-MDU compilation, guarded preflight, and execution.

1. Identify

Select the exact PR #18 CSV under assets/validated/.

2. Qualify

Recompute fixed A3 gates and require the approved registry result.

3. Compile

Build the high-level arm runner natively on the A3 MDU.

4. Preflight

Verify the package, approval, runtime, topics, and vendor state.

5. Prepare

Check hold behavior and move both arms to the committed READY pose.

6. Serve

Trigger the original fixed motion and restore vendor ownership afterward.

Important ownership boundary

The CSV contains every SDK joint column, but the validated high-level runtime publishes only the 14 arm joints. The installed A3 vendor stack retains control of the waist, legs, and neck.

Path	Purpose
assets/validated/	Exact PR #18 CSV, runtime contract, identity notes, and demo reference.
config/approved_motions.json	Reviewed approval record keyed to the PR #18 CSV SHA-256.
runtime/	Native-MDU C++ runner, protected wrapper, and package builder.
runtime/README.md	Detailed preflight, operator modes, handoff, and restoration behavior.
tests/	Identity, qualification, safety, and runtime-wrapper regression checks.

Application root: apps/a3_mujoco_serve/

02 / IMMUTABLE ASSET AND APPROVAL

Confirm the exact PR #18 motion

The relocated file is the same byte sequence as the machine-tested source. Qualification recomputes the existing A3 envelope and confirms that this exact hash is approved.

1

Byte-for-byte preserved

assets/validated/serve_policy.csv is identical to the PR #18 source. SHA-256: 2a7de3f1c97a300069899c139c9eb96e94fd61d3419701d5e44ef37b2bf6641d.

2

Git object identity also matches

The original and reorganized paths both resolve to Git blob 200a52552feb3ed5d978310fb2d95ca6150aff2e. Headers, values, formatting, and represented motion are unchanged.

3

Fixed-profile qualification is approved

The builder applies a3_high_level_arm_v1, matches the pinned approval registry, and reports the included CSV as approved. Approval does not bypass numerical validation.

Verify the default deployment input

```
bash runtime/scripts/build_a3_app.sh --verify-inputs-only
# expected: qualification: approved
# expected: input verification PASS; no executable was built
```

Evidence	Observed result
Motion identity	Original and relocated files share the same Git blob and SHA-256.
Safety profile	Source speed 5.1574 < 5.2 rad/s; command velocity ratio 0.31754 < 0.5.
Approval	Registry entry a3-pr18-serve-policy; qualification status approved.
Regression tests	33 passed, including identity, qualification, cleanup, and wrapper behavior.

03 / MDU BUILD AND PREFLIGHT

Rebuild the approved example on the A3 MDU

Run these commands on the robot's ARM64/aarch64 MDU. The default input is the exact PR #18 CSV and its reviewed approval record.

1

Open the application on the MDU

Use a complete HOPE checkout and source the installed vendor ROS 2 environment.

```
cd /path/to/HOPE/apps/a3_mujoco_serve
source /agibot/software/v0/entry/env/env.sh
```

2

Verify the included motion

Confirm the fixed profile and approval result before spending time on compilation.

```
bash runtime/scripts/build_a3_app.sh --verify-inputs-only
```

3

Build the default approved asset

The builder reruns qualification, compiles the runner, and copies the exact CSV without rewriting it.

```
bash runtime/scripts/build_a3_app.sh --jobs 2
```

4

Enter the isolated package and preflight

The no-argument wrapper verifies hashes, approval, executable, trajectory, vendor action, topics, and motion_player without publishing commands.

```
cd dist/a3_serve_vendor_arm
./run_a3_app.sh
```

Stop if preflight does not pass

Do not add the real-command confirmation flag to work around an error. Correct the package, qualification, vendor state, or topic ownership first.

04 / ON-MACHINE EXECUTION

Run the example motion in guarded stages

Every real mode requires an interactive terminal and the literal --confirm-real-commands argument. Use a trained operator, working e-stop, physical support, and a cleared workspace.

Stage	Command	Operator outcome
Preflight	./run_a3_app.sh	Read-only; requires package and approved qualification PASS.
Hold	./run_a3_app.sh --hold-only --confirm-real-commands	Hold measured arm state for three seconds.
Prepare	./run_a3_app.sh --prepare-only --confirm-real-commands	Move both arms to READY; hold until Ctrl-C.
Serve	./run_a3_app.sh --serve-only --confirm-real-commands	Wait for Space; play validated serve and return.

Recommended sequence

```
./run_a3_app.sh
./run_a3_app.sh --hold-only --confirm-real-commands
./run_a3_app.sh --prepare-only --confirm-real-commands
./run_a3_app.sh --serve-only --confirm-real-commands
```

REVERIFY

Rehash package, motion, qualification, and pinned registry at the runner boundary.

READY

Prearm from a fresh exact-name 14-joint state; no publisher is created yet.

TRIGGER

In serve-only mode, press Space at physical ball release.

RESTORE

Stop the custom publisher, then always attempt motion_player restoration.

Protected ownership handoff

The wrapper stops only motion_player through the vendor process manager, releases the prearmed runner after publisher ownership is clear, and restores motion_player after the custom publisher stops. Cleanup remains active after repeated signals or a dropped terminal.

Before every real run

Confirm e-stop reachability / safety support / clear arm and racket envelope / correct ball-release operator / no unexpected command publisher / MotionControlAction_MOTION is RUNNING.

05 / QUALIFICATION AND PREFLIGHT CONTRACT

What the package checks before motion

The default invocation is read-only. Real modes remain locked until the exact PR #18 asset, fixed safety profile, pinned approval registry, compiled runner, and robot state all agree.

Package identity

Verify every packaged file against the SHA-256 package manifest.

Motion identity

Match the exact CSV to source.sha256 in the motion manifest.

Qualification

Require fixed-profile PASS and approved registry status for real modes.

Executable

Require an AArch64 ELF with no unresolved shared libraries.

Trajectory

Validate finite values, A3 limits, frame count, steps, speed, and joint order.

Vendor and ROS

Require the expected motion action, idle player, topics, and guarded handoff.

Expected isolated package

```
dist/a3_serve_vendor_arm/
a3_serve_vendor_arm_runner run_a3_app.sh a3_app_common.sh
motions/serve_policy.csv
config/serve_vendor_arm_manifest.json
config/serve_vendor_arm_qualification.json
config/approved_motions.json
config/serve_vendor_arm_build.env
config/serve_vendor_arm_package.sha256
```

The preflight command is safe to repeat

Run ./run_a3_app.sh after a rebuild, package copy, vendor-stack restart, or robot reboot. It does not move the arms or replace motion_player's publisher.

Failure behavior

A missing approval, changed registry, hash mismatch, invalid trajectory, wrong architecture, unresolved dependency, unexpected robot state, or publisher conflict stops the wrapper before real commands.

06 / VERIFICATION AND REFERENCES

What was checked in this revised local build

The package was checked against the original PR #18 identity, fixed-profile qualification, pinned approval registry, runtime contracts, shell syntax, and portable regression tests.

Check	Observed result
Git blob	Original and relocated asset: 200a52552feb3ed5d978310fb2d95ca6150aff2e.
SHA-256	2a7de3f1c97a300069899c139c9eb96e94fd61d3419701d5e44ef37b2bf6641d.
Qualification	a3_high_level_arm_v1 PASS; status approved; approval ID a3-pr18-serve-policy.
Measured gates	Source speed 5.1574 < 5.2 rad/s; command velocity ratio 0.31754 < 0.5.
Runtime	Exact CSV copied unchanged; all real modes reverify package and approval.
Tests	33 passed; shell syntax and repository diff checks passed.

Primary technical references

Reference	Repository location
Application guide	apps/a3_mujoco_serve/README.md
Runtime guide	apps/a3_mujoco_serve/runtime/README.md
Validated assets	apps/a3_mujoco_serve/assets/validated/README.md
Approval registry	apps/a3_mujoco_serve/config/approved_motions.json
Qualification	apps/a3_mujoco_serve/a3_serve/qualification.py
Original motion	apps/a3_mujoco_serve/assets/validated/serve_policy.csv
Demo source	apps/a3_mujoco_serve/assets/validated/pr18_a3_serve_demo.mp4

Shortest approved-example reminder

```
cd /path/to/HOPE/apps/a3_mujoco_serve
source /agibot/software/v0/entry/env/env.sh
bash runtime/scripts/build_a3_app.sh --verify-inputs-only
bash runtime/scripts/build_a3_app.sh --jobs 2
cd dist/a3_serve_vendor_arm && ./run_a3_app.sh
./run_a3_app.sh --serve-only --confirm-real-commands
```

Use the full hold and prepare stages on first deployment or whenever the robot, vendor stack, package, workspace, or operator setup changes.